

Quiz Master App

(Modern Application Development –1)

Student Details

Name - Rehan Alam

Roll.no - 23f2002420

Email- 23f2002420@ds.study.iitm.ac.in

Description

This project is multi-user quiz preparation platform where an administrative(admin) manages subjects, chapters, quizzes and questions. The admin creates and manages quiz content, while user attempt quizzes and view their score.

Technology-used

1. Backend

- Flask: A Python web framework used to handle routing, user authentication and backend logic.
- SQL Alchemy: An Object-Relational Mapper used for interacting with sqlite3 database to manage the application's data models and relationships.
- SQLite: A self-contained database used to store all user, quiz, questions and administrative data.
- Matplotlib: A python library used to generate visual charts.

2. Frontend

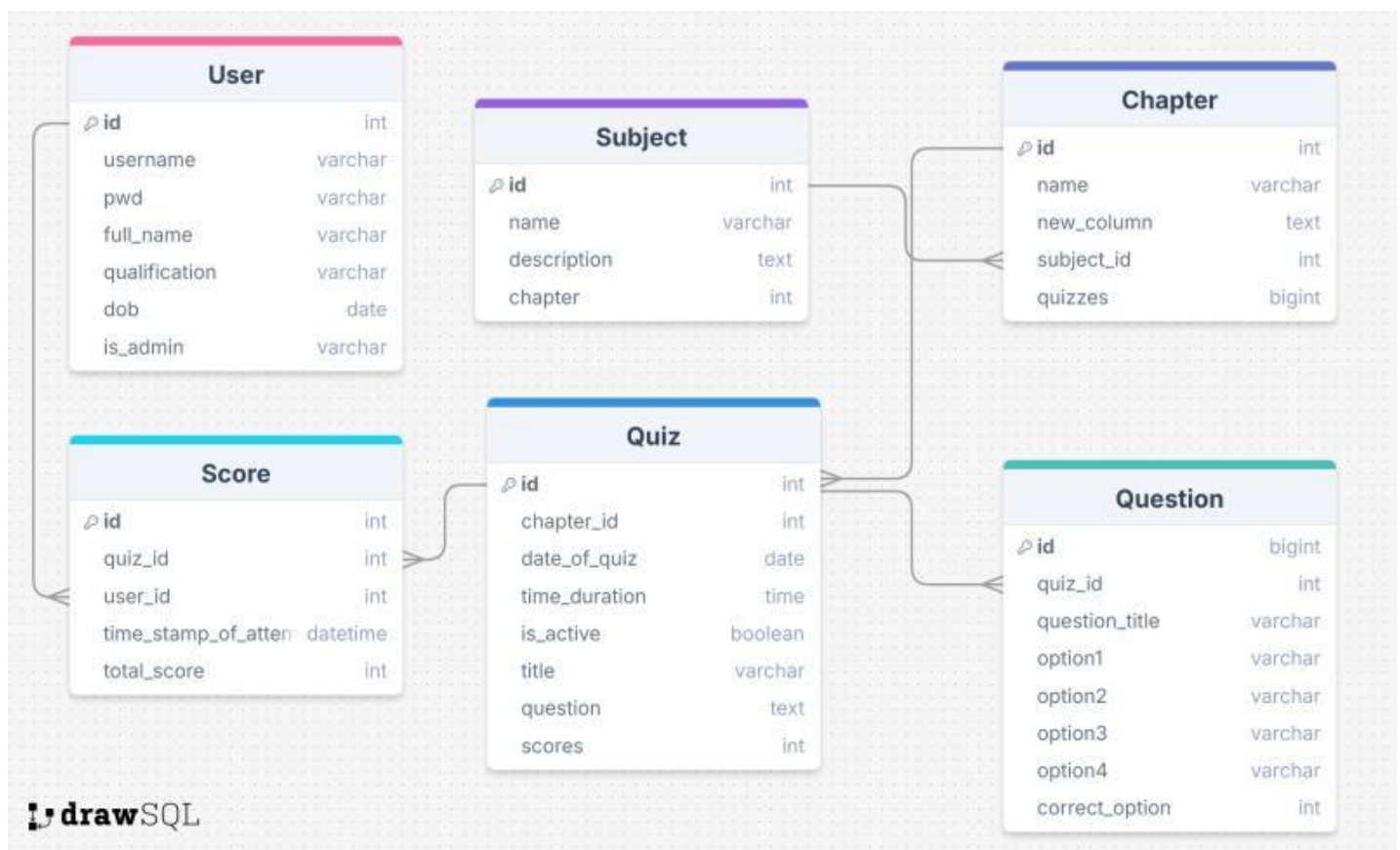
- HTML/CSS/Bootstrap: Used to design a responsive and user-friendly interface for both the admin and user dashboards.
- Jinja2: Template engine used to dynamically render HTML pages based on backend-data.

3. Additional Technology

- Python-3: The programming language used to develop the logic and backend functionality.
- Git: Version control to manage code changes and collaboration.
- Flask-session: Session management for storing user data across requests.
- Datetime: Utilized for handling and manipulating date and time, particularly in quiz timing and score tracking.

- pytz: A library for accurate time-zone management, ensuring that date and time calculations are precise regardless of user location.
- SQLAlchemy.func: Provides powerful functions for database operations, such as aggregation (e.g., sum, average) and other calculations.

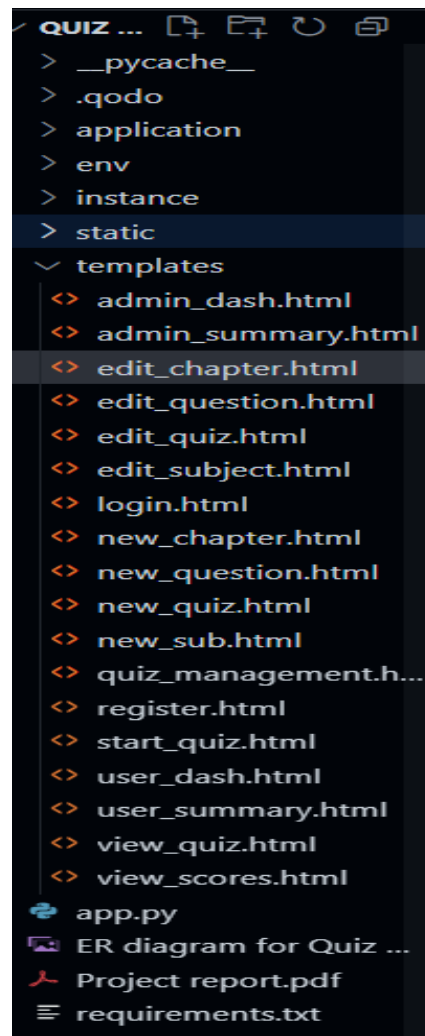
DB Schema Design



Architecture

Here, the app.py file contains the main code to run the app.

- Controllers.py: This file contains manages for backend.
- Models.py: This file contains all types of models how users, subjects, questions, quizzes, chapters are connected.
- Database.py: DB defined.
- Static: This contains CSS for styling.
- Templates: This contains all html templates for user admin, quizzes, questions, subjects, chapters, score, edit, login, register, summary.



Features:

The application provides a range of features for both the **Admin** and **User** roles, focusing on ease of use and interactivity. Below are the key features implemented:

1. Admin Dashboard:

- a. The **admin** (Quiz Master) has full control over the platform. They can create, edit, and delete subjects, chapters, and quizzes.
- b. Admins can also manage user accounts by viewing and deleting them if necessary.
- c. **Quiz Management:** Admins can add multiple MCQ questions under each quiz, specifying the correct answers and other quiz details.
- d. **Data Visualization:** The admin dashboard includes **Matplotlib**-generated charts that show quiz performance summaries and user statistics.

2. User Dashboard:

- a. Users can register, login, and view available quizzes for each subject and chapter.
- b. Users can start quizzes with a built-in **timer**, attempt MCQs, and receive scores after completing a quiz.
- c. **Previous Quiz History:** Users can view a summary of their past quiz attempts and scores.

3. User Registration & Login:

- a. Users can sign up by providing their personal information such as username, email, password, and qualification.
- b. Both users and the admin can securely log in to their respective dashboards.

4. Quiz Attempts:

- a. Users can select a quiz based on subjects and chapters, and attempt it in an interactive format.
- b. **Timer Functionality:** Each quiz comes with a timer (optional) to ensure users complete the quiz within the set duration.
- c. **Quiz Score Tracking:** Scores are saved in the database, allowing users to view their performance after completing the quiz.

5. Admin-User Interaction:

- a. The admin has the ability to view all users, their quiz history, and scores, helping them manage and track overall platform usage.
- b. Admins can add new questions, and manage quiz content by editing, updating, or deleting quizzes.

6. Additional Features:

- a. **Session Management:** Using **Flask-Session**, user sessions are managed to keep track of logged-in users and their ongoing quiz attempts.

b. **Time Zone Handling:** **Pytz** ensures that date and time information (such as quiz attempt timestamps) is accurate across different time zones.

Video Link